

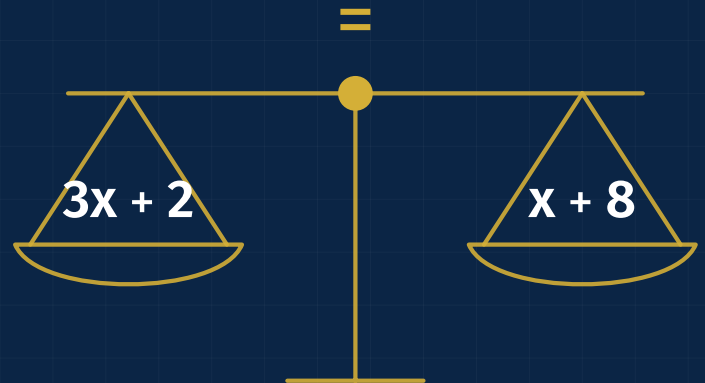


DR ESLAM AHMED
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CHAPTER 2 • ALGEBRA • TOPIC 13

Forming Equations

Turn the words or the picture into an equation, solve it — then answer the question they actually asked.



6

VARIANTS

5

PAST-PAPER QUESTIONS

53

PRACTICE QUESTIONS

YOU WILL MASTER

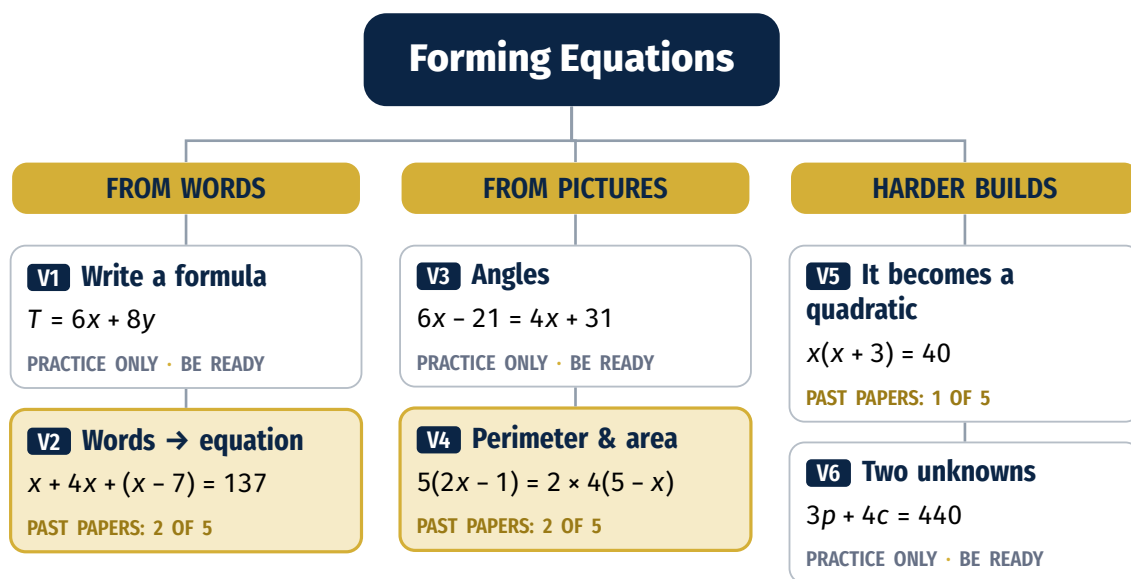
V1 Write a formula V2 Words → equation V3 Angles

V4 Perimeter & area V5 It becomes a quadratic V6 Two unknowns

TOPIC MAP · EVERY VARIANT ON ONE PAGE

Forming Equations at a glance



Your notes


THREE MOVES

LETTER → EQUATION → ANSWER

x is rarely the final answer
— read the last line again.

 How often does each variant come up?

5 past-paper questions (Jan 2020 → Nov 2025), 4–6 marks each; 53 practice parts — grouped by type.

	PAST PAPERS (OUT OF 5)		PRACTICE PARTS
V1 Write a formula	● ● ● ● ●	0	3
V2 Words → equation	● ● ● ● ●	2	8
V3 Angles	● ● ● ● ●	0	6
V4 Perimeter & area	● ● ● ● ●	2	10
V5 It becomes a quadratic	● ● ● ● ●	1	20
V6 Two unknowns	● ● ● ● ●	0	6

WHAT THIS TELLS YOU Every past-paper question ended with a **final step** after solving (“tin C”, “the zinc bars”, “L and W”). Solving is half the marks — answering is the rest.

BEFORE YOU START

The big idea



Your notes
Words are just algebra in disguise.

Use a letter for the unknown, translate each phrase with the **dictionary**, and look for the fact that says “**is**”, “**total**”, “**equals**” — that is your = sign.

the words

4 times the number
7 fewer than x
3 more than y
twice as old as Ben
half of n
the total is 137
 b is 3 more than $\sqrt{a}$

the algebra

$4x$
 $x - 7$
 $y + 3$
 $2b$
 $\frac{n}{2}$
 $\dots = 137$
 $b = \sqrt{a} + 3$

**The facts you will need****Triangle**

$a + b + c = 180^\circ$
angles

Point / quadrilateral

360°
angles

Rectangle

$P = 2(l + w)$, $A = lw$
perimeter, area

Right triangle

$a^2 + b^2 = c^2$
Pythagoras

**Dr Eslam's three moves****1 LETTER**

what is x ?

2 EQUATION

dictionary + the fact

3 SOLVE & ANSWER

the question asked

**CALCULATOR FIRST, THEN CHECK**

Check by putting the answer back into the story: 24 buttons in tin A $\rightarrow$ B has 96, C has 17 $\rightarrow$ total $24 + 96 + 17 = 137$ ✓. If the story works, your answer is right.

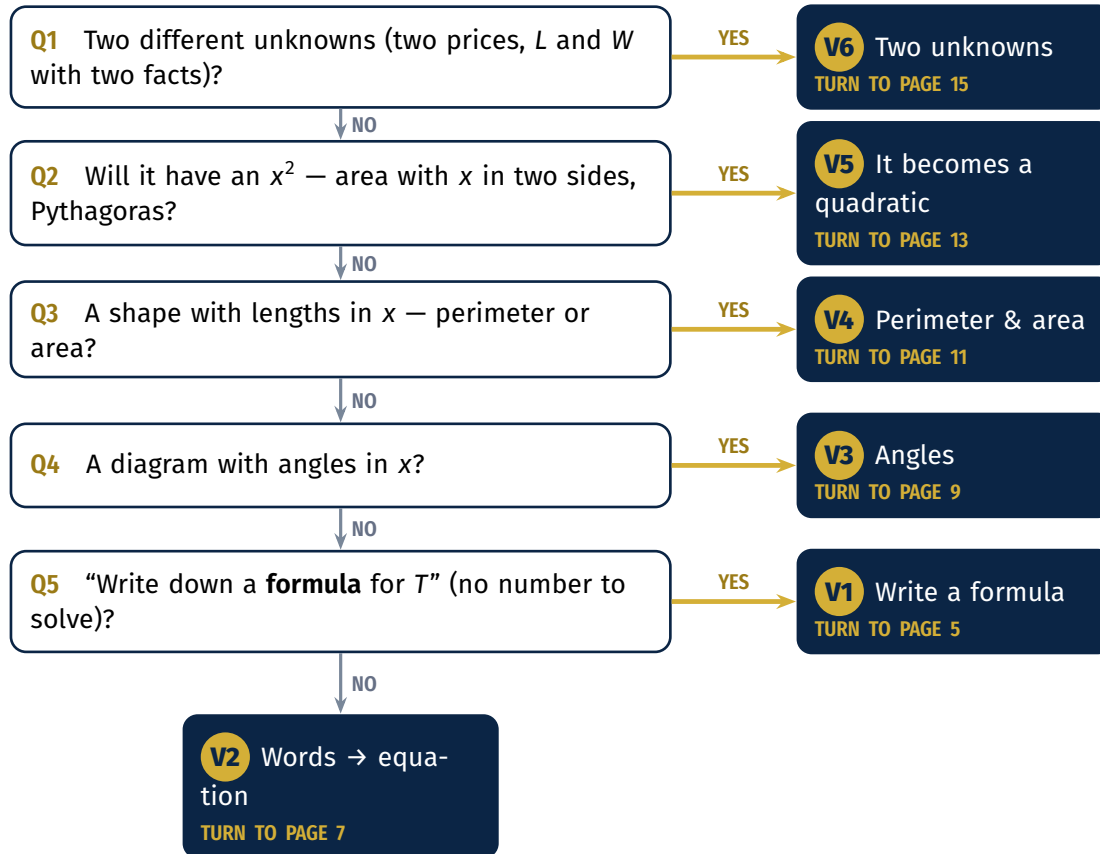
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Write a Formula

No number to solve — just build the expression for the total.

SEEN IN PAST PAPERS

●●●●● 0/5

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 3



Your notes

? What is it?

In plain words

“Write a formula for T ” means $T =$ an expression. Count **each group**: x packs of 6 buns is $6x$ buns.

A formula must start with $T =$ — an expression alone loses the mark.

6 6 6 x packs of 6 $\rightarrow 6x$

8 8 y boxes of 8 $\rightarrow 8y$

IT LOOKS LIKE

$$T = 6x + 8y$$

$$T = 5x + 20y$$

$$T = 6x + 8y$$

$$b = \sqrt{a} + 3$$

How do I do it?

1 Each group

Number in one group $\times$ how many groups.

2 Add the groups

The total is the sum of the groups.

3 Start with the subject

Write $T = \dots$ — not just the expression.

A FORMULA STARTS WITH $T =$ Check with small numbers.

V1 Write a Formula

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Write down a formula for T in terms of x and y " → $T = (\text{group 1}) + (\text{group 2})$

" b is 3 more than the square root of a " → $b = \sqrt{a} + 3$ (root first, then +3)

"Circle the correct equation" → Translate in order, then compare

WORKED EXAMPLE **PRACTICE**

Forming Equations · Easy Q2 · 3 marks

Kalinda buys x packs of currant buns and y boxes of iced buns. A pack has 6 buns, a box has 8. She buys T buns in total. Write a formula for T in terms of x and y .

STEP 1 $6x$ and $8y$ ◀ each group

STEP 2 $6x + 8y$ ◀ add the groups

STEP 3 $T = 6x + 8y$ ◀ start with $T =$

ANSWER $T = 6x + 8y$ ✓ $x = 1, y = 1$: 14 buns ✓**WATCH OUT**✗ $6x + 8y$ ✓ $T = 6x + 8y$

The question asks for a formula — the $T =$ is part of the answer.

EXAM TIP

Test it: 2 packs and 1 box should be $12 + 8 = 20$ buns — does your formula give 20?

Check yourself

calculator first · answers on page 17

1.1 Practice · Easy

Small packets hold 5 envelopes, large ones 20. Formula for T envelopes in x small and y large packets?

Answer:

1.2 Practice · Easy

b is 3 more than the square root of a . Write the equation.

Answer:

1.3 Dr Eslam

Pens cost 45p, rulers 30p. Write a formula for the cost C (pence) of x pens and y rulers.

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM**Words → Equation**

A story with one unknown: write each amount in x , the total gives the equation, then answer.

SEEN IN PAST PAPERS

● ● ● ● ● **2 / 5**

PRACTICE QUESTIONS: 8



Your notes

Came up in: Jan 2020 P2HR Q3 · Nov 2023 P1H Q2(b)

? What is it?**In plain words**

Write **every amount** in terms of the one letter. The **total** (or “is”, “the same as”) makes the equation.

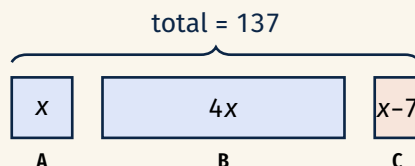
Solve — then do the **last step**: the question usually asks for a different amount, not x .

IT LOOKS LIKE

tins of buttons

metal bars

ages

**How do I do it?****1 Letter**The amount the others are built from: “Tin A contains x ”.**2 Every amount**Dictionary: $4x$, $x - 7$, $3y + 7$...**3 Equation & solve**

Add them = the total. Clean → isolate.

4 Answer the question

Substitute back to get the amount they asked for.

x IS NOT THE ANSWER Read the last line again.

V2 Words → Equation

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"4 times the number", "7 fewer than" → $4x$, $x - 7$

"The total number is 137" → **Add all the expressions = 137**

"Work out the number of buttons in tin C" → **Solve, then substitute into C's expression**

WORKED EXAMPLE **PAST PAPER**

January 2020 · Paper 2HR · Q3 · 4 marks

Tin A has x buttons. Tin B has 4 times as many as tin A. Tin C has 7 fewer than tin A. The total is 137. Work out the number of buttons in tin C.

STEP 2 $A = x$, $B = 4x$, $C = x - 7$ ◀ every amount in x

STEP 3 $x + 4x + x - 7 = 137 \Rightarrow 6x = 144$ ◀ the total, collect

STEP 3 $x = 24$ ◀ solve

STEP 4 $C = 24 - 7 = 17$ ◀ answer the question

ANSWER 17 buttons in tin C✓ $24 + 96 + 17 = 137$ ✓**WATCH OUT**

✗ answer: 24

✓ answer: 17

24 is tin A. The question asked for tin C.

EXAM TIP

Write the line " $C = x - 7 = 17$ ". The final mark is for answering what was asked.

Check yourself

calculator first · answers on page 17

2.1 Nov 2023 P1H

56 bars: y gold, $(3y + 7)$ silver, $(2y - 5)$ zinc. How many are zinc?

Answer:

2.2 Dr Eslam

I think of a number, multiply by 3, subtract 7: the answer is 20. Find the number.

Answer:

2.3 Practice · Medium

Gemma and Betty have the same number. Gemma gives Betty 24; Betty now has 5 times as many. Total?

Answer:

V3**VARIANT 3**
AnglesAngles in x : use an angle fact to write the equation.

SEEN IN PAST PAPERS

● ● ● ● ● 0/5

NOT ASKED YET · BE READY

PRACTICE QUESTIONS: 6


 Your notes
? What is it?**In plain words**

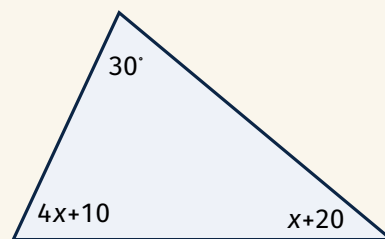
The angles are expressions in x . Pick the **angle fact** that links them: a triangle adds to 180° , a point to 360° , equal angles are equal.

Solve for x , then work out the angle they asked for.

IT LOOKS LIKE

$$30 + (4x + 10) + (x + 20) = 180$$

$$6x - 21 = 4x + 31$$

sum = 180° **How do I do it?****1 The fact**

Triangle 180° · point 360° · straight line 180° · equal angles.

2 Equation

Add (or set equal) the angle expressions.

3 Solve

Collect and divide.

4 The angle

Substitute x to find the angle asked for. Check the sum.

NAME THE ANGLE FACT

It earns a mark on its own.

V3 Angles

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

A triangle → **Angles add to 180°**

"Angle AOC = angle BOC" → **Set the two expressions equal**

"Is AC a straight line?" → **Show the angles add to exactly 180°**

**WORKED EXAMPLE** **PRACTICE**

Forming Equations · Easy Q4 · 3 marks

Angle AOB is a right angle. Angle AOC = $(6x - 21)^\circ$ and angle BOC = $(4x + 31)^\circ$ are equal. Work out x .

STEP 1 $\angle AOC = \angle BOC$ ◀ the fact: they are equal

STEP 2 $6x - 21 = 4x + 31$ ◀ equation

STEP 3 $2x = 52 \Rightarrow x = 26$ ◀ solve

STEP 4 $135^\circ + 135^\circ + 90^\circ = 360^\circ \checkmark$ ◀ check round the point

ANSWER $x = 26$ ✓ both angles 135° **WATCH OUT**

✗ adding to 180 without a reason

✓ choose the right fact first

Equal angles → set them equal; triangle
→ 180; point → 360.

**EXAM TIP**

Always check: put x back, add the angles
— they must give 180 or 360.

**Check yourself**

calculator first · answers on page 17

3.1

Practice · Easy

A triangle has angles 30° , $(4x + 10)^\circ$ and $(x + 20)^\circ$. Find x .

Answer:

3.2

Dr Eslam

A triangle has angles x° , $2x^\circ$, $3x^\circ$. Find the largest angle.

Answer:

3.3

Dr Eslam

A quadrilateral has angles x , $2x$, $3x$, $4x$. Find the largest angle.

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Perimeter & Area

Lengths in x : perimeter or area gives the equation.

SEEN IN PAST PAPERS

●●●●● **2 / 5**

PRACTICE QUESTIONS: 10



Your notes

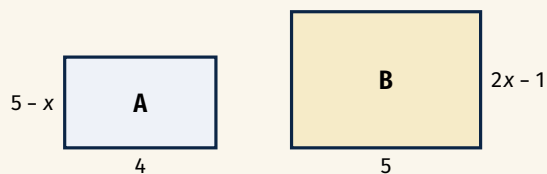
Came up in: Nov 2021 P2H Q2 · Nov 2021 P1H Q5

? What is it?

In plain words

Perimeter: add **all** the sides (a rectangle has 4!).**Area:** length $\times$ width — with brackets.“B is twice A”: **area B = $2 \times$ area A**.

Two shapes the same: set them equal.



$$\text{area B} = 2 \times \text{area A}$$

IT LOOKS LIKE

$$4x + 8 = 45$$

$$5(2x - 1) = 2 \times 4(5 - x)$$

How do I do it?

1 Write the measures

Area or perimeter of each shape in x (brackets!).

2 The fact

“Perimeter is 45”, “twice the area”, “the same”.

3 Solve

Expand, collect, divide.

4 Answer

The length / area they asked for — with units.

BRACKETS ROUND EVERY SIDE $5(2x - 1)$, not $5 \times 2x - 1$.

V4 Perimeter & Area

VARIANT 4 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"The perimeter is 45 cm" → **Add all the sides = 45**

"The area of B is twice the area of A" → **Area B = 2 × area A**

Tiles making a big rectangle → **Read the big sides from the tiles:**
 $5W = 4L$

WORKED EXAMPLE PAST PAPER

November 2021 · Paper 2H · Q2 · 4 marks

Rectangle A is 4 cm by $(5 - x)$ cm. Rectangle B is 5 cm by $(2x - 1)$ cm. The area of B is twice the area of A. Work out x .

STEP 1 area A = $4(5 - x)$, area B = $5(2x - 1)$ ◀ the measures

STEP 2 $5(2x - 1) = 2 \times 4(5 - x)$ ◀ the fact

STEP 3 $10x - 5 = 40 - 8x \Rightarrow 18x = 45$ ◀ expand, collect

STEP 4 $x = 2.5$ ◀ solve

ANSWER $x = 2.5$

✓ A = 10, B = 20: twice ✓

WATCH OUT

× 2× on the wrong side:

$$2 \times 5(2x - 1) = 4(5 - x)$$

✓ area B = 2× area A

Read it as a sentence: "B is twice A" →
 $B = 2A$.

EXAM TIP

Check with your x : work out both areas
 — is one really twice the other?

Check yourself

calculator first · answers on page 17

4.1 Practice · Easy

A rectangle is x cm by $(x + 4)$ cm with perimeter 45 cm. Find x .

Answer:

4.2 Nov 2021 P1H

9 tiles L by W make a rectangle with $5W = 4L$ and area 1620 cm^2 . Find L and W.

Answer:

4.3 Practice · Medium

Trapezium: $AD = BC = x$,
 $AB = 2x$, $DC = 2x + 4$,
 perimeter 38 cm. Find AD.

Answer:

V5

VARIANT 5

It Becomes a Quadratic

x times x (area, Pythagoras): ERCS, solve, reject the impossible root.

SEEN IN PAST PAPERS

● ● ● ● ● 1 / 5

PRACTICE QUESTIONS: 20



Your notes

Came up in: Nov 2025 P2H Q20

? What is it?**In plain words**

When x is in **two** sides that multiply (area) or are squared (Pythagoras), you get x^2 — a quadratic.

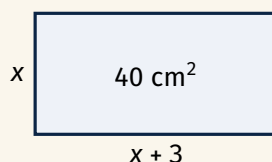
ERCS to = 0, solve (Topic 10), and **reject** a negative length.

IT LOOKS LIKE

$$x(x + 3) = 40$$

$$x^2 + (x + 7)^2 = 13^2$$

$$5w^2 = 1620$$



$$x(x + 3) = 40$$

$$x^2 + 3x - 40 = 0$$

How do I do it?**1 Build**

Area or Pythagoras with the expressions (brackets).

2 ERCS

Expand, rearrange to = 0, collect.

3 Solve

Calculator → factorise or the formula.

4 Reject

A length cannot be negative — say so.

REJECT THE NEGATIVE LENGTH

...and say why.

V5 It Becomes a Quadratic

VARIANT 5 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DOArea with x in two sides→ **A quadratic: ERCS, then solve**A right-angled triangle with x sides→ **Pythagoras: $a^2 + b^2 = c^2$** "Show that $x^2 + 3x - 40 = 0$ "→ **Start from the formula, end exactly there****WORKED EXAMPLE DR ESAM**

Area • 4 marks

A rectangle is x cm wide and $(x+3)$ cm long. Its area is 40 cm^2 . (a) Show that $x^2 + 3x - 40 = 0$. (b) Find the width.

STEP 1 $x(x+3) = 40$

◀ (a) area = length $\times$ width

STEP 2 $x^2 + 3x - 40 = 0$

◀ (a) expand, rearrange

STEP 3 $(x+8)(x-5) = 0 \Rightarrow x = 5 \text{ or } x = -8$

◀ (b) calculator roots $\rightarrow$ brackets

STEP 4 $x = 5$ ($x = -8$: a width can't be negative)

◀ reject

ANSWER width 5 cm✓ $5 \times 8 = 40$ ✓**WATCH OUT**✗ " $x = 5 \text{ or } x = -8$ "✓ " $x = 5$; -8 rejected (a length)"

The question asks for one width — give one, and say why the other goes.

EXAM TIP

In "show that", write each line from the formula to the given equation — never start from the answer.

Check yourself

calculator first • answers on page 17

5.1 Practice • Easy

A rectangle's length is 5 times its width; area 1620 cm^2 . Find the width.

Answer:

5.2 Nov 2025 P2H

$ax^2 + 4x + c = 0$ has solutions $x = \frac{-4 \pm 2\sqrt{39}}{10}$. Find a and c .

Answer:

5.3 Dr Eslam

A right-angled triangle has sides x , $(x+7)$ and hypotenuse 13 cm. Find x .

Answer:

V6

VARIANT 6

VARIANT 6

Two things you don't know, two facts: two equations — simultaneous.

SEEN IN PAST PAPERS

0 / 5

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 6



Your notes

What is it?

In plain words

Two prices (or two lengths) are unknown. Use **two letters**, and each fact gives **one equation**. Solve them together (Topic 12).

Shortcut: if the question asks for $p + c$, adding the equations may give it straight away.

$$\begin{array}{r} 3p + 4c = 440 \\ + \\ 4p + 3c = 470 \\ \hline 7p + 7c = 910 \end{array} \quad p + c = 130$$

IT LOOKS LIKE

$$3p + 4c = 440 \text{ \& } 4p + 3c = 470$$

How do I do it?

1 Two letters

Say what each means: “ $p = 1$ kg of potatoes (pence)”.

2 Two equations

One per fact, same units.

3 Solve

Eliminate (Topic 12) — or add/subtract if it gives what is asked.

4 Answer

The amount asked for, with units (£ or p).

TWO UNKNOWN → TWO EQUATIONS Then Topic 12.

V6 Two Unknowns

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"3 kg of potatoes and 4 kg of carrots cost 440p" → $3p + 4c = 440$

"Find the total cost of 1 kg of each" → **Try adding the equations**

"half as many boys as girls" → $2b = g$

WORKED EXAMPLE **PRACTICE**

Forming Equations · Medium Q6 · 4 marks

3 kg of potatoes and 4 kg of carrots cost 440p. 4 kg of potatoes and 3 kg of carrots cost 470p. Work out the total cost of 1 kg of potatoes and 1 kg of carrots.

STEP 2 $3p + 4c = 440$, $4p + 3c = 470$ ◀ two letters, two equations

STEP 3 $7p + 7c = 910$ ◀ add them

STEP 3 $p + c = 130$ ◀ divide by 7

STEP 4 $130p = £1.30$ ◀ answer with units

ANSWER £1.30✓ $p = 80$, $c = 50$: $240 + 200 = 440$ ✓**WATCH OUT**

× mixing pence and pounds

✓ all in pence (or all in £)

440p and £1.30 in the same equation gives nonsense.

EXAM TIP

Look at what is asked before solving fully — sometimes one add or subtract gives it.

Check yourself

calculator first · answers on page 17

6.1 Dr Eslam

3 pens and 2 pencils cost £2.60; 1 pen and 2 pencils cost £1.40. Find each price.

Answer:

6.2 Dr Eslam

Tiles are L by W cm.
 $2L + W = 11$ and
 $L + W = 7$. Find the area of one tile.

Answer:

6.3 Dr Eslam

A class of 30 has 6 more girls than boys. How many of each?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py). Got one wrong? Go back to that variant's Watch-out box first.

1.1 $T = 5x + 20y$

1.2 $b = \sqrt{a} + 3$

1.3 $C = 45x + 30y$

2.1 $y = 9$: 13 zinc bars

2.2 $x = 9$

2.3 72 sweets

3.1 $x = 24$

3.2 90°

3.3 144°

4.1 $x = 9.25$

4.2 $L = 15, W = 12$

4.3 $AD = \frac{17}{3} \approx 5.67 \text{ cm}$

5.1 18 cm

5.2 $a = 5, c = -7$

5.3 $x = 5$

6.1 pen 60p, pencil 40p

6.2 12 cm^2

6.3 12 boys, 18 girls

THE WHOLE TOPIC ON ONE PAGE

Forming Equations cheat sheet



Your notes
V1 Write a formula

$$T = 6x + 8y$$

Each group, add them, start with $T =$.

V2 Words → equation 2/5

$$x + 4x + (x - 7) = 137$$

Every amount in x ; total $=$; answer the question.

V3 Angles

$$6x - 21 = 4x + 31$$

Name the angle fact; check the sum.

V4 Perimeter & area 2/5

$$5(2x - 1) = 2 \times 4(5 - x)$$

Brackets round sides; "twice" goes on the smaller.

V5 Quadratic 1/5

$$x(x + 3) = 40$$

ERCS, solve, reject the negative length.

V6 Two unknowns

$$3p + 4c = 440$$

Two facts, two equations, Topic 12.

**TOP 3 LOST MARKS**

1. stopping at x (not the asked amount) $\times$
2. "twice" on the wrong side $\times$
3. keeping a negative length $\times$

**THE DICTIONARY**

4 times $\rightarrow 4x$ · 7 fewer $\rightarrow x - 7$ · twice as old $\rightarrow 2b$ · half $\rightarrow \frac{n}{2}$ · total / is $\rightarrow =$

BOTTOM LINE Letter $\rightarrow$ equation (dictionary + the fact) $\rightarrow$ solve $\rightarrow$ answer the question they asked.